Description

Illustration





Figure caption

1. Close-up of the material. © ANSYS, Inc. 2. The Basilica of Pisa. © ANSYS, Inc.

The material

Marble is the purest form of limestone. It is almost pure calcium carbonate that has been compacted and heated such that it has recrystallized. It is used for ornamental building, statuary, ornamental furniture and for electric power panels. The purest marble (Italian Carrara marble is an example) is very white, delicate in texture and hard. Other marbles are dark green, red, black or gray, allowing their use for decorative patterning like that of the Basilica shown in the picture.

Compositional summary

Calcium carbonate (CaCO3).

General properties

Density		2.72e3	-	2.85e3	kg/m^3
Price	*	0.31	-	0.785	GBP/kg
Date first used		-10000			

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✓
Services	✓

Mechanical properties

Young's modulus		50	-	70	GPa
Shear modulus	*	22	-	28	GPa
Bulk modulus	*	26	-	36	GPa
Bending modulus	*	45	-	80	GPa
Poisson's ratio		0.14	-	0.22	
Yield strength (elastic limit)		6	-	10	MPa
Tensile strength		6	-	10	MPa
Compressive strength	*	55	-	105	MPa
Bending strength	*	7	-	11	MPa
Elongation	*	0.01	-	0.02	% strain
Hardness - Vickers		16	-	20	HV
Fatigue strength at 10^7 cycles	*	5	-	8	MPa
Fracture toughness		0.6	-	1.2	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	5e-4	-	0.001	

Thermal and combustion properties

Thermal conductor or insulator?		Poor insulator			
Thermal resistivity		0.167	-	0.2	m.°C/W
Thermal expansion coefficient		3	-	5	µstrain/°C
Specific heat capacity		850	-	890	J/kg.°C
Melting point		1.23e3	-	1.34e3	°C
Maximum service temperature	*	630	-	680	°C
Minimum service temperature		-273			°C
Flammability		Non-flammable			
Emissivity		0.75	-	0.95	

Hygro-thermal properties

Water absorption		0.1	-	0.5	%
Water vapor permeability		4.73e-13	-	6.51e-12	kg.m/s.m^2.Pa
Frost resistance		Good			

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity	*	1e8	-	1e12	µohm.cm
Dielectric constant (relative permittivity)	*	6	-	8	
Dissipation factor (dielectric loss tangent)	*	5e-4	-	0.001	
Dielectric strength (dielectric breakdown)	*	5	-	12	MV/m

Optical properties

Natural color

Pure white, dark green, red, black or gray.

Transparency	Opaque	
Transmissivity	0	%

Critical Materials Risk

High critical material risk?	No

Acoustic properties

Sound absorption	Poor
Sound isolation	Good

Processability (scale 1 = impractical to 5 = excellent)

Machinability	3	-	4	

Durability

Water (fresh)	Excellent
Water (salt)	Excellent
Weak acids	Unacceptable
Strong acids	Unacceptable
Weak alkalis	Excellent
Strong alkalis	Acceptable
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Wear resistance	Limited use
Industrial atmosphere	Limited use
Rural atmosphere	Excellent
Marine atmosphere	Excellent

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	*	0.162	-	0.179	kg/kg
Embodied energy (virgin grade production)	*	2.53	-	2.79	MJ/kg
Water Usage	*	3.23	-	3.57	l/kg

Material processing: energy

Machining, grinding, embodied energy	*	20.5	-	28	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO_2-eq)	*	1.34	-	1.83	kg/kg

Recycling and end of life

Recycle	✗			
Functional recycle	✗			
Recycle fraction in current supply	1.34	-	1.48	%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Design guidelines

Marble, nearly pure calcium carbonate, is easily cut and carved. Its fine grain size makes it ideal for detailed carving. Marble is hard and dense, and takes a near-perfect polish. It has a wonderful translucency, making it the choice of many sculptors. Like limestone, it weathers well in a clean environment but is attacked by acid, and thus by industrial emissions.

Technical notes

Good marble is expensive. Marblelite is an artificial substitute used for casting statues and small objects. Crushed marble is used as an aggregate in flooring.

Typical uses

Buildings, facings, floors, stairs, statuary and ornamental furniture, electric-power panels.

Related standards

04400 Stone
04410 Rough Stone
04420 Cut Stone
04440 Flagstone
04450 Stone Veneer
04455 Marble

Links

ProcessUniverse
Reference

Values marked * are estimates.
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